TASK 1

Pick the partner edge and justify SOAP

Choose one external partner **CampusEats** must integrate — a payment gateway, a logistics/delivery partner, or an SMS gateway. In 3–4 sentences, explain why that edge is **SOAP** while the rest of **CampusEats** is **REST** (strong contract, message-level security, transactions).

- **Partner Edge**: GlobalPay External Financial Gateway Integrated
- **Operation**: Charge
- **Context & Justification**: **CampusEats** integrates with GlobalPay, an external banking and payment gateway, to handle direct card charges for users.
- **Explanation** – **CampusEats** microservices here is considered a collection of lightweight **REST APIs** for operational agility. The edge connection to the external **GlobalPay payment gateway** is considered a **SOAP** because to satisfy strict enterprise financial standards. Financial institutions demand rigid, strongly typed **WSDL contracts** to eliminate **data specification ambiguity during monetary transactions**. Additionally, SOAP natively provides **enterprise-grade WS-Security for message-level encryption and strict ACID transaction** guarantees across external organizational boundaries.

TASK 4

Show the HTTP binding

Show the **HTTP POST** that carries the request: **method, Host, Content-Type, the SOAPAction** value from your **binding**, and the **endpoint URL** from your **service/port**.

- HTTP POST request transmitting the SOAP payload to the payment gateway:

```
POST /soap/v1/charge HTTP/1.1
Host: api.globalpay.com
Content-Type: text/xml; charset=utf-8
Content-Length: 728

SOAPAction: "http://campuseats.org/paymentgateway/Charge"
<?xml version="1.0" encoding="UTF-8"?>

<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
 xmlns:pay="http://campuseats.org/paymentgateway">
            <soapenv:Header>
                  <pay:AuthHeader>
                        <pay:merchantId>CAMPUS_EATS_001</pay:merchantId>
                        <pay:apiKey>sk_live_9f8d7c6b5a4e3d2c1</pay:apiKey>
                  </pay:AuthHeader>
            </soapenv:Header>
 <soapenv:Body>
      <pay:ChargeRequest>
          <pay:studentId>usr_883920194</pay:studentId>
          <pay:orderId>ord_551029481</pay:orderId>
          <pay:amountCents>1450</pay:amountCents>
```

```
        <pay:paymentMethodId>pm_card_visa_4242</pay:paymentMethodId>
    </pay:ChargeRequest>
 </soapenv:Body>
</soapenv:Envelope>
```

TASK 5

Describe discovery (modern form)

Explain how **CampusEats** would obtain this **WSDL** — a direct **URL**, or a **lookup** — and write a **one-record catalogue / registry entry**: **business**, **service**, **endpoint**, and the **WSDL** it speaks (a **tModel-style pointer**). No live **UDDI** server is required.

- Rather than hosting a legacy **UDDI server**, **CampusEats** utilizes a modern internal **API Service Catalogue (Registry)**. During deployment, the Payment Service fetches the formal partner contract directly via the partner's HTTPS metadata URL endpoint **("https://api.globalpay.com/soap/v1/charge?wsdl")**

Service Registry Entry (tModel-Style Pointer):

Field	Value
Business Name	**GlobalPay Financial Services Ltd.**
Service Name	**CampusEats** External Payment Processing Gateway
Service ID	srv-ext-payment-globalpay-v1

Transport Protocol	SOAP 1.1 over HTTP/1.1
Target Endpoint URL	https://api.globalpay.com/soap/v1/charge
WSDL Pointer (tModel)	https://api.globalpay.com/soap/v1/charge?wsdl
Contract Target Namespace	http://campuseats.org/paymentgateway
Primary Operation	**Charge**

TASK 6

Map the fault into your own contract

Show how the **partner's fault** becomes the error your **Assignment 2** contract promised (e.g. the partner's card_declined → your placeOrder "payment declined"). The partner's vocabulary must not leak to your students.

- To prevent **external vendor terminology** from leaking to **student clients or internal interfaces**, the **internal Payment Service** intercepts and translates **SOAP XML Faults** into standard **CampusEats** internal **contract** errors.

External SOAP Fault Code (GlobalPay Gateway)	External Fault Detail (Partner Vocabulary)	Internal Contract (Error Ouput)
ERR_INSUFFICIENT_FUNDS	Card balance unavailable	**PAYMENT_DECLINED**
ERR_CARD_EXPIRED	Expiry date passed	**PAYMENT_DECLINED**
ERR_AUTH_FAILED	Invalid Merchant Credentials	**SERVICE_UNAVAILABLE**